

Sentinel

A Fully Procedural Game Engine
and Open-World Military Shooter

*Nothing is loaded from a model, image or sound file:
everything is generated from code and a seed.*

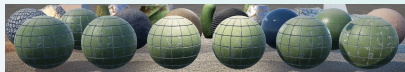
Md. Shifat Hasan

Roll 2107067 · 4th Year 1st Term

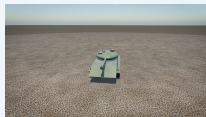
Instructors: Md Tajmilur Rahman, Md. Muftashim Abrar Nihal
Computer Graphics and Image Processing Laboratory · October 7, 2026

Outline

1. Problem, objectives, architecture
2. Procedural shapes and textures
3. Transformations: creating and combining objects
4. Illumination models and light sources
5. Rendering pipeline: shadows, ambient, tone map
6. Results, verification, a bug fixed
7. Video tour of the project



Six illumination models



1. Problem, objectives and architecture

Can a complete, playable 3D game be made by code alone on a student laptop?

TWO HARD LIMITS

No model, image or precomputed file;
OpenGL 4.1, no compute shaders.

OBJECTIVES

Layered engine; all content from code; modern renderer; a playable game (vehicles, ladders, stealth, five-objective mission); tested, 16.7 ms budget.

Game: mission, AI, vehicles

Render: deferred, lights, sky

World: collision, terrain

Procedural: noise, textures

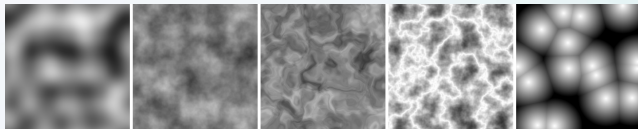
GPU: shaders, buffers

Platform: window, input



A lower layer never knows a higher one.

2. Procedural shapes and textures



gradient, fractal, warped, ridged and cellular noise

FRACTAL NOISE (TILES EXACTLY)

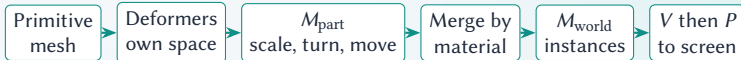
$$F(\mathbf{x}) = \frac{\sum_{i=0}^{n-1} 2^{-i} v(2^i \mathbf{x})}{\sum_{i=0}^{n-1} 2^{-i}}$$

VERTEX PUSH

$$\mathbf{p}' = \mathbf{p} + A v(f\mathbf{p}) \hat{n}, \quad \|\mathbf{p}' - \mathbf{p}\| \leq A$$

- 18 recipes $\rightarrow$ 22 materials, baked once at 1024^2
- **Scharr** filter turns height into bumps; crevices darken
- Objects: tables of primitives and noise deformers
- Soldiers: skeletons with two-bone IK legs

3. Transformations: creating and combining objects



ONE PART: SCALE, TURN, THEN MOVE

$$M_{\text{part}} = T(\mathbf{t}) R_z(\gamma) R_y(\beta) R_x(\alpha) S(\mathbf{s})$$

Right to left: stretched, turned about its own centre, moved last. $(1, 0, 0) \rightarrow S(2, 1, 1) \rightarrow R_y 90^\circ \rightarrow T(5, 0, 0) = (5, 0, -2)$

NORMALS USE THE INVERSE TRANSPOSE

$$\hat{n}' = \frac{(L^{-1})^\top \hat{n}}{\|(L^{-1})^\top \hat{n}\|}$$

$S = \text{diag}(2, 1, 1)$: correct $(0.447, 0.894, 0)$,
naive $(0.894, 0.447, 0)$

ROTATION ABOUT y

$$R_y(\theta) = \begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix}$$

HIERARCHY: PRODUCTS DOWN THE CHAIN

$$M_{\text{gun}} = M_{\text{hull}} T(\mathbf{c}_t) R_y(\theta_t) T(\mathbf{c}_g) R_x(-\theta_g)$$

WORLD TO SCREEN

$$\mathbf{p}_{\text{clip}} = P V M_{\text{world}} M_{\text{part}} \mathbf{p}$$

4. Illumination models and light sources

DIRECT LIGHT + AMBIENT + EMISSION

$$L_o = \sum_j f(\mathbf{l}_j, \mathbf{v}) L_j \text{ att}_j \text{ sh}_j \max(0, \mathbf{n} \cdot \mathbf{l}_j) + \text{ambient} \cdot \text{AO} + \mathbf{e}$$

SIX ILLUMINATION MODELS

Lambert	$f = \frac{c_d}{\pi}$
Phong	$f = \frac{c_d(1 - F_0)}{\pi} + F_0 \frac{m + 2}{2\pi} (\mathbf{r} \cdot \mathbf{v})^m$
Blinn-Phong	$f = \frac{c_d(1 - F_0)}{\pi} + F_0 \frac{m + 8}{8\pi} (\mathbf{n} \cdot \mathbf{h})^m$
Oren-Nayar	$f = \frac{c_d}{\pi} (A + B \max(0, \cos \Delta\varphi) \sin \alpha \tan \beta)$
Cook-Torrance	$f = \frac{(1 - F_l)(1 - F_v) c_d}{\pi} + D V F$
Subsurface	$\cos \theta \rightarrow \frac{\mathbf{n} \cdot \mathbf{l} + w}{1 + w}, \quad w = 0.5$

LIGHT SOURCES

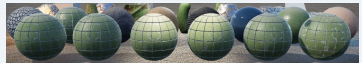
Directional: sun, moon

Point: lamps, flashes

Spot: floodlights

Area: 4×4 samples

$$\text{att}(d) = \frac{\text{clamp}(1 - (d/R)^4, 0, 1)^2}{d^2 + 1}$$



Six models, one sphere each

5. Shadows, ambient light and tone mapping



Deferred shading: store colour, normal, shininess and depth first; light each pixel once.

CASCADED SHADOW MAPS

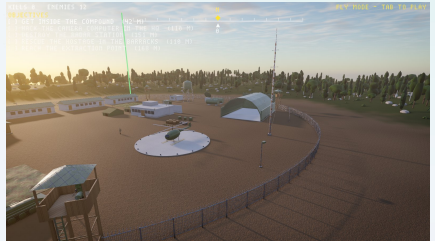
$$z_i = \lambda n \left(\frac{r}{n} \right)^{i/N} + (1 - \lambda) \left(n + (r - n) \frac{i}{N} \right)$$

$\lambda = 0.75$, $N = 3$: cuts at 9.1, 24.2, 100 m;
normal-offset lookup, 8 rotated PCF taps.

FOG AND TONE MAP

$$C = T C_s + (1 - T) C_h, \quad T = e^{-kd}$$

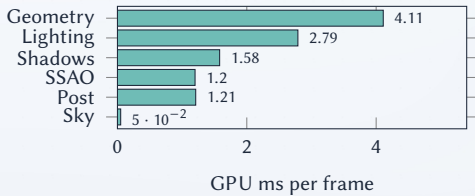
$$C_{\text{out}} = \frac{x(2.51x + 0.03)}{x(2.43x + 0.59) + 0.14}$$



Dusk: sun, sky, haze, shadows

Ambient = sky diffuse + split-sum specular, times SSAO.

6. Results, verification and a bug fixed



10.9 ms of the 16.7 ms budget at 1080p

Unit tests (13 packages)	281
TextureCheck (exact tiling)	8 318
RenderCheck (lighting probes)	318 567
GpuCheck	30
Playthroughs, 5/5 objectives	2

Bug: a vertical line at night. Fog sampled the whole sky, so one horizon star streaked a column; fog now uses the smooth atmosphere only.



before



after

LIMITS AND FUTURE WORK

Blocky models, no true global illumination, lumpy foliage. Next: leaf cards, screen-space GI, point-light shadows, multiplayer.

7. Video tour of the project

References

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Thank you

Questions are welcome
Md. Shifat Hasan · Roll 2107067